

ProcureWise

Vendor Comparison & Procurement Assistant

Capstone topic	04 — Vendor Comparison and Procurement Assistant
Track	Gen AI for Business
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Repository	github.com/udaykondeti/muj · branch: claude/full-project-7yb6fd · /Capstone

Summary

ProcureWise is a working web application that turns messy vendor quotations (arriving as emails, PDFs and spreadsheets) into a standardised, transparently scored comparison and a **human-approved** procurement recommendation. A deterministic engine handles the maths and compliance rules so every number is reproducible and auditable; generative AI (Anthropic Claude) reads the messy quotes, explains the recommendation in plain language, drafts vendor emails and answers procurement-policy questions. The system deliberately never places an order — a named human signs off every award.

Problem / Opportunity

Buyers compare inconsistent quotes by hand, miss compliance requirements such as security certification or warranty minimums, and rarely record *why* a vendor was chosen. The opportunity is to standardise quotes automatically, score them reproducibly against weighted criteria and policy, and keep an auditable, human-approved decision — cutting sourcing cycle time while improving transparency and governance.

Solution overview — one working journey

- **Standardise:** paste any quote; AI extracts it into one comparable structure.
- **Score:** a deterministic engine normalises each criterion, applies the buyer's weights, computes Total Cost of Ownership and ranks vendors.
- **Gate on compliance:** a vendor failing a mandatory policy check (e.g. no ISO 27001) is never recommended, even if cheapest.
- **Explain:** AI writes a plain-language rationale from the scoring table.
- **Review & decide:** a named human approves, overrides or rejects; the system cannot transact.
- **Communicate & audit:** AI drafts clarification emails and grounded policy answers; every decision is logged and an exportable memo is produced.

Where Gen AI adds value vs. deterministic logic

This separation is deliberate. AI is used for language; the numbers, rules and the purchase decision are not left to AI.

Concern	Handled by
Reading messy free-text quotes	Gen AI (Claude)
Explaining the recommendation	Gen AI
Drafting vendor clarification emails	Gen AI
Procurement-policy Q&A (grounded)	Gen AI
Normalisation, TCO, weighted score, ranking	Deterministic engine
Mandatory compliance gating	Deterministic engine
Placing an order / financial commitment	Nobody — a human approves

Key deliverables

- Working application (zero-dependency Node.js server + vanilla single-page dashboard).
- Deterministic scoring engine with 16 passing unit tests (determinism, bounds, compliance gating).
- Gen AI module (Claude) with a deterministic fallback so the app runs with no API key.
- Automated Playwright walkthrough and 9 screenshots of the full journey.
- Exportable Markdown decision memo; architecture notes; this overview.

Business metrics the solution could improve

- **Sourcing cycle time** — hours of manual comparison reduced to minutes.
- **Realised savings** — quantified per award (4.9% under budget in the demo).
- **Compliance adherence** — mandatory checks are enforced, not optional.
- **Audit completeness** — 100% of awards carry a named approver and rationale.

Limitations: data is synthetic; production would need vendor master data, authentication, and a database. The system is decision-support only and cannot make purchases.

Screenshots

ProcureWise
PROCUREMENT
ASSISTANT

Overview

Quotations

Criteria & Weights

Comparison

Recommendation & Review

Clarification Emails

Policy Q&A

Audit Trail

AI: deterministic mode

Reset demo

NorthBridge Retail Group

Field-Team Laptop Fleet · 150 units

Comparison

Deterministic, auditable scoring across every criterion.

Full comparison matrix

Per-criterion normalised score (0–100) × weight. Highlighted row = recommended.

Export decision memo

#	VENDOR	UNIT PRICE 30%	QUALITY RATING 20%	DELIVERY TIME 15%	WARRANTY 15%	SUPPORT RESPONSE SLA 10%	PAYMENT TERMS 10%	TOTAL	COMPLIANCE
1	BlueByte Technologies ₹1,02,75,000 TCO	₹68,500 0→0	4.6/5 100→20	14 days 100→15	36 months 100→15	8 hours 100→10	45 days 100→10	70	✓ pass
2	Apex Computing Solutions ₹93,45,000 TCO	₹62,000 75.6→22.67	4.2/5 50→10	21 days 56.3→8.44	24 months 50→7.5	24 hours 60→6	30 days 50→5	59.6	✓ pass
3	CoreVantage Systems ₹90,45,000 TCO	₹59,900 100→30	3.8/5 0→0	30 days 0→0	12 months 0→0	48 hours 0→0	15 days 0→0	30	✗ ISO 27001 information-security certified

Each cell shows the raw value, then **normalised→weighted** contribution. The total is the sum of weighted contributions — fully reproducible.

Comparison matrix — each cell shows raw value then normalised→weighted contribution, so every score is reproducible. CoreVantage is blocked for missing ISO 27001.

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Recommendation & Review

AI explains; a named human approves. The system never buys.

AI recommendation summary

Generate explanation

deterministic

Recommendation: BlueByte Technologies

BlueByte Technologies ranks #1 with a weighted score of **70/100** for the 150-unit field-team laptop fleet.

- Strongest on:** Quality rating (100/100), Delivery time (100/100), Warranty (100/100).
- Fleet cost (TCO):** INR 1,02,75,000 — about INR 68,500 per unit including freight.
- Estimated saving vs the INR 72,000/unit budget:** INR 5,25,000 (4.9%).
- Compliance:** passes all mandatory checks (100% of the full checklist).
- Excluded from award:** CoreVantage Systems — ISO 27001 information-security certified.

Compared with Apex Computing Solutions (59.6) and CoreVantage Systems (30), BlueByte Technologies offers the best balance of the weighted criteria while meeting every mandatory policy requirement.

This is a decision-support summary generated from the scoring table. ProcureWise does not place orders — a named human approver must sign off before any commitment.

Human decision (required)

ProcureWise cannot place an order. Record an accountable human decision below.

Award to

BlueByte Technologies — score 70

Decision

Approve recommendation

Approver name

e.g. R. Menon, Category Manager

Rationale

Why this decision? (kept in the audit trail)

Record decision

No PO is issued — this only records the human sign-off.

Key numbers

	BlueByte Technologies
Recommended	
Score	70/100
Fleet TCO	₹1,02,75,000
Saving vs budget	₹5,25,000 (4.9%)
Saving vs costliest quote	₹0 (0%)

Recommendation & Review — AI rationale (left) built from the numbers; the mandatory human decision (right). The system never places an order.